CRITERION – II	
KEY INDICATOR	**2.6 Student Performance and Learning Outcomes**
METRIC NO	**2.6.1**

Programme Outcomes (POs) and Course Outcomes (COs) for all Programmes

CHRIST COLLEGE PUNE

DEPARTMENT OF ARTS

Programme Outcome

After the successful completion of three-year BA (Economics) Programme, the graduate will be able to:

PO1: Demonstrate disciplinary knowledge by defining, describing, and discussing major theories and concepts in Digital Humanities and literature through academic engagements. (Graduate Attributes: Academic Excellence, Domain Expertise)

PO2: Demonstrate critical thinking skills through analysing various texts and contexts and research skills by designing and executing independent research project/s with the aid of various digital tools. (Graduate Attributes: Academic Skills & Professional Excellence)

PO3: Display effective communication and social interaction skills in a multimodal manner and adapt to an ever-changing digital landscape through collaborative assignments, internships, and developing industry networks. (Graduate Attributes: Interpersonal skills/Leadership/Communication)

PO4: Cultivate responsible citizenship, display environmental awareness and ethical conduct to address local, regional, national, and global needs through community engagements. (Graduate Attributes: Societal/ Social and Environmental Sensitivity, Cross-Cultural Recognition, Civic Sense, Respecting Diversity, Personal skills)

Bachelor of Arts (Economics)

Semester I

Course Code	Course Title	Course Outcome
11001	**Compulsory English**	1.Demonstrate reading, writing, speaking, and listening skills. 2.Develop the rules of grammar. 3. Discover personality by instilling core values and life skills. 4. Adapt soft skills.
11371	**Sociology: Introduction to Sociology**	1. Understand the social context of emergence of Sociology. 2. Build basic sociological concepts and subject matter and perspectives of Sociology 3. Estimate the new avenues in Sociology
11161A	**Politics: Introduction to Indian Constitution**	1. Learn and discover the important features of the Constitution of India and with the basic framework of Indian Government. 2. Discuss the working of the Constitution of India.
11171	**Early India: From Prehistory to the Age of the Mauryas**	1. Compare the history of early India from the prehistoric times to the age of the Mauryas. 2. Construct the factors and forces behind the rise, growth and spread of civilization and culture of India along with the dynastic history.

		3. Acquaint the students about the contribution of Early Indians to polity, art, literature, philosophy, religion and science and technology. 4. Imagine the spirit of enquiry among the students by studying the major developments in early Indian history.
11151	**Economics: G-1 Indian Economic Environment**	1. Ability to develop an understanding of the economic environment 2. Interpret the various new developments in the different sectors of an economy – agriculture, industry, services, banking, etc 3. Compare and contrast Indian Economy with other world economies. 4. Discuss and debate on the 3 various issues and challenges facing the Indian Economic Environment.
11331	**Optional English**	1. Identify the basics of English literature and language. 2. Infer the minor forms of literature and use the language creatively. 3. Improve the pronunciation skills. 4. Build their communication skills.

Semester II

Course Code	Course Title	Course Outcome
12331	**Optional English**	1. Identify the basics of English literature and language. 2. Understand the minor forms of literature and use the language creatively. 3. Improve their pronunciation skills. 4. Enhance their communication skills.
12171	**History: Early India: Post Mauryan Age to the Rashtrakutas**	1. Illustrate the history of India after the Mauryas along with the developments which finally led to the transition to medieval India. 2. Examine the developments in different parts of India through a brief study of regional kingdoms up to the tenth century C.E. 3. Analyse the consequences of the foreign invasions, particularly on the polity, economy, society and art and architecture.
12161	**Politics: Introduction to Indian Constitution**	1. Outline the important features and working of the Indian Constitution. 2. Examine the relation between Rights, Duties and DPSP in the Constitution of India. 3. Analyse the functioning of Legislature, Executive and Judiciary. 4. Compare various constitutional provisions in the Indian Constitution.

12151	**Economics: G-1 Indian Economic Environment**	1. Define the role and growth of service sector in Indian economy 2. Classify the challenges faced by the Indian Service sector 3. Examine the current business scenario, agricultural scenario and other sectorial growth in the Indian context. 4. Compile the developments such as MSME's, Digital Economy, E-Banking, BPO, KPO etc.
12371	**Sociology: Social Institutions and Change**	1. Understand the inter-relationship between Macroeconomic aggregates 2. Acquaint themselves to various aggregates of national income and its inter-relationships and the dynamic functioning of an economy. 3. Develop the critical view on the recent fiscal policy of Indian Economy and how that affect Employment and Output. 4. Analyse and demonstrate knowledge of the theories in Macroeconomics 5. Practically apply macroeconomic concepts, models and its use in real life situations.
12001	**Compulsory English**	1. Classify reading, writing, speaking, and listening skills. 2. Understand the rules of grammar. 3. Develop personality by instilling core values and life skills. 4. Adapt soft skills.

Semester III

Course Code	Course Title	Course Outcome
23152	**Macro Economics**	1. Introduce students to the historical background of the emergence of macroeconomics 2. Familiarize students with the differences between microeconomics and macroeconomics 3. Elaborate with various concepts of national income 4. Construct with Keynesian macroeconomic theoretical framework of consumption and investment functions 5. Examine the role of money in an economy. 6. Analyse the conceptual and theoretical frameworks of inflation, deflation and stagflation, Business Cycle.
23163	**An Introduction to Political Science**	1. Learn and infer the important sub themes of Political Science as a discipline 2. Outline the Approaches to study Political Science 3. To relate with the Basic Concepts and Values in Political Science
23151	**Micro Economics**	1. Develop an understanding about subject matter of Economics 2. Impart knowledge of Micro Economics and clarify Micro Economics concepts 3. Analyse and interpret charts, graphs and figures 4. Design the application of basic economic theories.
		1. Demonstrate his/her understanding of sampling methods and the ability to use collection of data

32154	**Basic Concepts of Research Methodology**	2. Develop appropriate techniques for different research questions. 3. Discover the appropriate source of data in relation to collection of research data 4. Assess and present the collected data in the form of graph, bar diagram, chart etc
23153	**Financial System**	1. Build the fundamentals of modern financial system in our country as well as the international financial institutions operating in the global economy. 2. Discover the present Era is of huge opportunities, challenges and development in every sector. 3. Formulate the important changes taking place in the Indian financial sector.
233333	**Advanced Study of English Language**	1. Identify the various components of language. 2. Categorise their linguistic skills in their future prospects and research. 3. Detect pronunciation errors and improve their pronunciation skills 4. Analyse the phonological, morphological, semantic and syntactical areas of language study. 5. Discuss the working of English language as a communicative medium and thereby enhance their communicative skills.
23999	**AEC Environmental Science**	1. Gain in depth knowledge on Natural processes to sustain life. 2. Predict the consequences of human actions on the web of life. 3. Develop Critical thinking for shaping strategies. 4.Acquire Value and Attitudes towards understanding.

23001	**Compulsory English**	1. Examine knowledge of key concepts, theories, and facts relevant to the subject matter. 2. Analyze and evaluate information and arguments critically. 3. interpret areas of personal growth and development related to the course. 4. Improve the creativity in problem-solving or project work.

Semester IV

Course Code	Course Title	Course Outcome
24333	**Advanced Study of English Language**	1. Identify the various components of language. 2. Display their linguistic skills in their future prospects and research. 3. Detect pronunciation errors and improve their pronunciation skills 4. Analyse the phonological, morphological, semantic and syntactical areas of language study. 5. Formulate the working of English language as a communicative medium and thereby enhance their communicative skills.
24163	**Politics: CC 1C Introduction to Political Science**	1. Demonstrate the important sub themes of Political Science as a discipline. 2. Discuss the Approaches to study Political Science. 3. Compare with the Basic Concepts, Values and Ideologies in Political Science. 4. Adapt the role of various International Organizations in Political Science.

24153	**Economics: CC-1C Financial System**	1. Define the fundamentals of Modern Financial System 2. Discover the structure and functioning of Commercial and Co-operative Banking in India 3. Analyze Indian Financial System into Indian Money Market, Indian Capital Market and Indian Foreign Exchange Market 4. Evaluate the changing landscape of Banking sector in India
24151	**Micro Economics**	1. Demonstrate the concept of revenue and its application 2. Classify the various market structures prevalent in the economy and examine the pricing policies 3. Examine the issues in the factor market both from demand and supply side and provide comprehensive solutions 4. Compare and analyse the factor incomes that accrue to factors of production 5. Evaluate the criteria for welfare in an economy and interpret the welfare aspect in different economies
24154	**Basic Concept of Research Methodology**	1. Develop data analytics skills and meaningful interpretation to the data sets so as to solve the business/Research problem 2. To solve numerical questions using simple statistical tools 3. Explain the basic concept of qualitative research techniques
24001	**Compulsory English**	1. Apply critical thinking skills to solve complex problems within the course's domain. 2. Construct quantitative methods and analytical tools, as appropriate to the subject matter.
24152	**Macro Economics**	1. Apply the basic concepts of macroeconomics, including GDP, inflation, unemployment, and economic growth. 2. Contrast the significance of GDP as a measure of economic performance.

		3. Examine the components of aggregate expenditure, including consumption, investment, government spending, and net exports. 4. Analyze the relationship between inflation and unemployment.
24999	**Environmental Science**	1. Develop the scope and importance of environmental science. 2. Discover the strategies for controlling pollution in air, water, and soil. 3. Analyse the legal framework for environmental protection. 4. Compare the different energy sources, including fossil fuels, renewable energy, and nuclear power.

Semester V

Course Code	Course Title	Course Outcome
35163	**Modern Political Analysis**	1. Demonstrate the study of Modern Political Analysis and power. 2. Build the nature of power in modern societies- more in the form of organizations and social formations than as individual power. 3. Compare the different forms of justifications of power and the role of ideology in this regard. 4. Interpret the repository of power in society and how the nature of power is shaped by social factors.
GR9A	**Financial Literacy**	1. Determine and set personal financial goals. 2. Discover the importance and implement the process of personal budgeting. 3. Estimate the importance of and overcome barriers to successful savings strategies using Mutual fund investments. 4. Formulate the basic concept of Mutual Funds and the types.

		5. Explain the role and importance of risk management from a personal finance perspective 6. Create the role and responsibilities associated with financial planning as a career
35152	**Public finance**	1. Relate and recognize the nature and scope of Public Finance 2. Compare and analyse the concept of Public Revenue and its components 3. Evaluate the types of Public Expenditure and reasons for its increase 4. Discuss the types of Public Debts and its effects
35151	**International Economics**	1. Demonstrate the Role and Scope of International Economics, Role of Trade and its impact at the national and international level. 2. Construct the concepts and role of international trade theories. 3. Compare alternative theories of International Trade and apply them in real world scenarios. 4. Examine the terms of trade in international markets and factors affecting the terms of trade. 5. Analyse nation's Balance of payments policies, Evaluate the impact of Tariff and non-Tariff barriers. 6. Assess how a foreign exchange market operates, identify the validity and efficiency of protectionist policies. 7. Create the opportunities and risks for India viz-a-viz global economies and also understand the different economic crisis and their global impacts.
		1. Demonstrate the differences between Economic growth and Development, Indicators of Economic Development. 2. Compare the Characteristics of Developing Countries. 3. Criticize the constraints on Development Process.

35153	**Economic Development**	4. Evaluate theories and Approaches of economic development. 5. Analyse some growth models 6. Formulate the macroeconomic policies, roll of foreign capital and economic planning etc. in developing countries.
35001	**Compulsory English**	1. Infer and identify different career opportunities. 2. Develop competence in using English for the career. 3. Discover skills required for placements and use them effectively. 4.Design the verbal and non–verbal communication effectively.
35154	**Business Management - I**	1. Demonstrate the theory and practice of management and its development phases till date. 2. Discover, nourish and nurture managerial traits and talents among the students. 3. Classify the business management and its development through the functions of planning, organizing, staffing, leadership and control. 4. Develop the insights about harmonizing individual and group goals with organisational goals by demonstrating organisational and professional leadership skills

Semester VI

Course Code	Course Title	Course Outcome
36333	**Enhancing Employability Skills**	1. Demonstrate the career opportunities available to them. 2. Identify the career opportunities suitable to them. 3. Develop the use of English in different careers. 4. Examine the competence in using English for the career of their choice. 5. Evaluate the skills required for their placement. 6. Formulate the verbal as well as nonverbal communication effectively for their career
36163	**Modern Political Analysis**	1. Define the Modern Political Analysis and Power. 2. Demonstrate the dynamics of Political Systems and Political Culture. 3. Examine the role of Political Socialization, Political Participation, Political Communication and Political Elite in different societies. 4. Analyse state as a repository of power, influence in society and how the nature of power is shaped by various social factors.
36152	**Public Finance II**	1. Explain and assess the components and instruments of Fiscal Policy 2. Identify the objectives of fiscal policy. 3. Relate to the concepts of Budget and its components 3. Compare and analyze the concept of Deficit Financing and its effects 4. Assess the Centre and State Financial Relationship 5. Evaluate the items of current and capital account of government budget.
		1. Compare pros and cons of different trade policies 2. Discuss the different exchange rate regimes and cite the facts about India's Trade Policy in recent years

36151	**International Economics - II**	3. Examine the changes in Import Export policies at national level and summarize the different steps taken towards promoting domestic exports 4. Outline the motive and role of different organizations working for international co-operation 5. Develop a case study of trade patterns by collecting data sets and by referring in established models of small and large nations.
35153	**Indian Economic Development II**	1. Infer and explain the process of economic planning 2. Discover the economic planning and developing goals 3. Examine and describe the changing structure of the planning process in India 4. Estimate the relation between Economic Development and Environment
36001	**Compulsory English**	1. Demonstrate and identify different career opportunities. 2. Develop competence in using English for the career. 3. Analyse skills required for placements and use them effectively. 4.Compare verbal and non–verbal communication effectively.
36154	**Business Management - II**	1. Interpret and apply strategic planning concepts, including the development of mission statements, vision statements, and strategic goals for organizations. 2. Adapt the fundamentals of project management, including project planning, scheduling, and execution.

CHRIST COLLEGE PUNE

DEPARTMENT OF COMMERCE

Programme Outcome

After the successful completion of three-year BCOM Programme, the graduate will be able to:

PO1: Construct the solutions for business challenges by applying the knowledge of financial analytics. (Professional, Academic)

PO2: Design innovative investment and auditing strategies to meet specific financial goals of clients in the changing global financial markets. (Professional, Communication, Leadership)

PO3: Acquire necessary professional knowledge and skills in banking, insurance and taxation adhering to the necessary ethical standards. (Professional, Personality)

PO4: Develop subject-related and transferable skills that are relevant in, higher education, employment and overall personality development. (Personality)

PO4: Enhance constructive decision-making ability through critical thinking, effective communication and collaborative efforts in professional environment. (Professional, Communication, Leadership)

PO5: Compare and contrast the conventional and contemporary business trends to develop innovative business models synchronized with ethics and social responsibility. (Social sensitivity, professional)

Bachelor of Commerce

Semester I

Course Code	Course Title	Course Outcome
112	**Financial Accounting- I**	1. Demonstrate the knowledge of basic accounting concepts 2. Examine the awareness about application of these concepts in business world 3. Assume the Knowledge about process of dissolution of partnership firm. 4. Design and finalization of accounts of various establishments.
113	**Business Economics (Micro) - I**	1. Infer knowledge of business economics 2: Discover the micro economic concepts. 3. Analyse and interpret charts and graphs. 4. Develop the basic theories, concepts of micro economics and their application
114A	**Business Mathematics & Statistics- I**	1. Apply the calculations in based on simple interest, compound interest, Percentages, ratio, proportion etc. 2. Determine the different statistical tools in practical applications.
		1. Understand evolution of banking. Outline structure of Indian Banking 2. Possess in-depth understanding of primary and secondary functions of a bank.

115B	**Banking & Finance- I**	3. Illustrate the concepts related to lending and ratios. 4. Examine the process of opening and operating procedure of bank accounts. Explain various types of bank accounts holders 5. Classify the conceptual clarity on various methods of remittance. 6. Estimate and provide knowledge of fundamentals of Banking 7. Create awareness about various banking concepts 8. Design banking operations
111	**Compulsory English**	1. Illustrate the reading, Writing, Speaking and Listening Skills. 2. Examine the certain rules of grammar 3. Develop personality by instilling core values & life skills.
117A	**Additional English**	1. Understand the exquisiteness and communicative power of English language 2. Instil universal human values through best pieces of literature in English 3. Formulate the literary terms and enhance their English communication skills
116C	**Marketing & Salesmanship**	1. Construct the marketing concepts, principles, and theories. 2. Determine the effective market research to gather consumer preferences and market trends. 3. Design the digital marketing channels and tools (e.g., social media, SEO).

Semester II

Course Code	Course Title	Course Outcome
121	**Compulsory English- II**	1. Identify literary techniques and creative uses of language in literary texts. 2. Adapt their texts to particular audiences and purposes. 3. Explain the relevance of themes found in literary texts to contemporary, personal, and cultural values. 4. Discover genres and conventions and demonstrate the awareness of the evolution theory of language by varied culture 5. Determine a platform to express creativity and talent, ignite critical thinking, and judge a work of art. 6. Adapt relevant theoretical concepts to literary or other texts and practices.
122	**Financial Accounting- II**	1. Demonstrate the Accounting Software Tally ERP 9. 2. Identify the business implications of financial statement information of NGOs' and other nonprofit organizations. 3. Elaborate the Intangible Assets and its valuation
123	**Business Economics- II**	1. Demonstrate an understanding of advanced economic concepts such as market structures, production and cost analysis, business strategy, and macroeconomic factors. 2. Ouline economic theory to solve business problems and make informed decisions.

		3. Estimate their economic analysis effectively, both in writing and verbally. 4. Develop and analyze complex business problems, evaluate different options, and make informed decisions
124A	**Business Mathematics and Statistics-II**	1. Define the basics of business mathematics and statistics 2. Predict the nature of growth in business world 3. Calculate the cost of living 4. Categorize various data
125B	**Banking And Finance – II**	1. Explain and define all the banking principles 2. Compare traditional and modern banking 3. Understand and use the various technologies in banking
126C	**Marketing & Salesmanship – II**	1. Remembering: Recall and recognize key concepts, terminology, and theories related to marketing and salesmanship. 2. Explain and summarize the principles and strategies of marketing and salesmanship. 3. Apply marketing and salesmanship techniques and theories in practical situations. 4. Analyze the effectiveness of various marketing and sales strategies and tactics. 5. Assess and evaluate the impact and success of marketing and sales campaigns and strategies. 6. Generate and design innovative marketing and sales plans and campaigns that meet the needs of specific target audiences.
		1.Understand the exquisiteness and communicative power of English language. 2. Instil universal human values through best pieces of literature in English.

127A	**Additional English - II**	3. Assess the literary terms and enhance their English communication skills. 4. Design the job potential of students by improving their language skills.

Semester III

Course Code	Course Title	Course Outcome
232	**Corporate Accounting -I**	1. Construct the financial statements of company within the frame work of accounting standard 2. Develop analytical skills in Trail balance, profit and loss account and balance sheets, stocks and debtors' system and final accounts system 3. Acquire the basic knowledge of the terms such as, departmental accounts, inter departmental transfer, branch accounting, stock and debtors system, depreciation, Valuation of shares.
233	**Business Economics (Macro)**	1. Understand the basic theories and concepts of Macro Economics and their application. 2.Compare the relationship between broad aggregates. 3. Interpret the knowledge of business economics. 4. Evaluate the macro-economic concepts.

		5. Elaborate the various concepts of National Income.
234	**Business Management -I**	1. Articulate the theory and practice of management and its development phases till date. 2. Discover, nourish and nurture managerial traits and talents among the students. 3. Design the business management and its development through the functions of planning, organizing, staffing, leadership and control.
235	**Elements of Company Law**	1. Acquaint with knowledge and maturity to understand Company law 2013. 2. Infer knowledge and application of formation and incorporation of Company. 3. Evaluate the principal documents of the company. 4. Discuss and inculcate skills and knowledge about the share capital of the company.
236E	**Cost and Works Accounting -I (Basics Of Cost Accounting)**	1. Understand and explain the conceptual framework of Cost and Works Accounting. 2. Equip to seek suitable career in Cost and Works Accounting and Entrepreneurship. 3. Construct the communication and analytical skill among students through self-learning. 4. Assess how to prepare a cost sheet, tenders and quotations. 5. Develop the various methods of Material, Labour and overheads control Techniques.

299AECC1	**Environmental Science**	1. Gain in depth knowledge on Natural processes to sustain life. 2. Predict the consequences of human actions on the web of life. 3. Develop Critical thinking for shaping strategies. 4. Acquire Value and Attitudes towards understanding.
231	**Business Communication - I**	1. Understand the basic principles of effective communication. 2. Examine strong written communication skills for emails, reports, and memos. 3. Evaluate the strategies for building positive workplace relationships. 4. Develop the use of technology tools for virtual meetings, webinars, and remote collaboration.

Semester IV

Course Code	Course Title	Course Outcome
241	**Business Communication – II**	1. Explain and summarize the principles of effective communication in business contexts, and demonstrate understanding of how they are applied in various situations. 2. Apply effective communication techniques in practical business situations, such as writing emails, creating presentations, and participating in meetings. 3. Analyze and evaluate the effectiveness of various types of communication used in business, such as written and oral communication, and identify areas for improvement.

		4.Assess and evaluate the impact of communication on business outcomes, such as customer satisfaction, employee engagement, and organizational success
242	**Corporate Accounting – II**	1. Observation and recall of all basic accounting concepts, principles, and terminology related to corporate accounting, such as the holding company, absorption, liquidation and forensic . 2. Explain the purpose and importance of corporate accounting, as well as the role of financial statements in providing information to stakeholders and comprehend how accounting transactions affect financial statements. 3. Apply accounting principles and techniques to prepare and analyze financial statements for a corporation. They should also be able to use accounting software and other tools to process accounting data and generate reports. 4. Analysis and evaluate the financial performance and position of a corporation using various ratios and other financial analysis techniques. They should also be able to identify and analyze accounting issues related to corporate transactions. 5. Assess the impact of accounting policies and practices on financial statements and the decision-making process of stakeholders.
244	**Business Management- II**	1. Gain comprehensive understanding of key business concepts, including finance, marketing, operations management, and organizational behavior. 2. Develop the ability to think critically and make informed decisions in a business setting. They will learn how to analyze complex business problems and develop effective solutions

		3. Outline leadership skills that will help them manage teams, motivate employees, and drive organizational success. They will also learn how to effectively manage change and navigate challenges in the workplace. 4. Formulate their communication skills, both written and verbal, which are essential for success in the business world. They will also learn how to present their ideas effectively to different audiences
245	**Elements Of Company Law – II**	1. Describe what is management of company, board of directors, their selection, remuneration & retirement. 2. Understand the KMP (key managerial personnel), how they arrive at decision. 3. Learn & apply how meetings are conducted & agenda are prepared & how minutes are written. 4. Analyze how E-governance works in practice & how company get winds up.
246E	**Cost And Works Accounting – II**	1. Understand the concept of the valuation of closing and opening stock. 2. Interpret the documents that are used in stores and how to calculate the issuing price of material. 3. Describe the concepts of labor turnover and merit rating.

		4. Compare the traditional and recent trends in cost accounting.
243	**Macro Economics**	1. Explain the inter-relationship between Macroeconomic aggregates 2. Understand the various aggregates of national income, its interrelationships and the dynamic functioning of an economy 3. Apply the Income, Employment and Output determination theories and its relevance in recent times and significance of fiscal policy 4. Analyse and demonstrate knowledge of the basic theories/laws in Macroeconomics 5. Evaluate macroeconomic concepts, models and its use in real life situations.
249AECC2	**Environmental Science**	1. Gain in depth knowledge on Natural processes to sustain life. 2. Predict the consequences of human actions on the web of life. 3. Develop Critical thinking for shaping strategies. 4. Design the Value and Attitudes towards understanding.

Semester V

Course Code	Course Title	Course Outcome
351	**Business Regulatory Framework**	1. Demonstrate conceptual knowledge about the framework of business Law in India. 2. Assess the legal aspect of business. 3. Create awareness among the students about legal environment relating to the Contract Law, Partnership Act, Sale of Goods Act in India. 4. Discover the emerging issues relating to e-commerce, e-transaction issues And E Contracts 5. Determine the career opportunity in corporate sector relating to business law in India 6. Adapt the concepts, terms; provisions of Mercantile and Business Laws.
352	**Advanced Accounting – I**	1. Demonstrate the various concepts, objectives, and applicability of some important accounting standards. 2. Assess the reorganization of business regarding restructuring the capital. 3. Inspect final accounts of a Banking Companies with the provisions of Banking Regulation Act 1949. 4. Estimate the skills to prepare the investment account in simple and summarized manner.
353	**International Economics-I**	1. Contrast the basic theories of International Trade and Economics. 2. Evaluate the working and functions of International Organisations and Institutions. 3. Develop a foundation in the subject that will help the students in their future academic and Professional Ventures

354	**Auditing**	1. Outline with knowledge and maturity to understand the concept of Auditing, types of Audits and Audit Process. 2. Apply Conceptual Clarity and Practical understanding of Vouching Verification and valuation and Types of Audit Report. 3. Discover the Practical knowledge about appointment, reappointment and other related provision. Practical knowledge about Tax Audit as per I.T. Act 1961 (Form 3CA, 3CB & 3CD). 4. Elaborate the new concepts under Audit of Computerized Systems & Forensic Audit.
355E	**Cost and Works Accounting. Special Paper II**	1. Infer the concept Overheads. 2. identify the Cost Accounting Standards. 3. Interpret the accounting treatment for under and over absorption of overheads 4. Discover the various methods of allocation of Overheads 5. Estimate the Activity Based Costing and to create awareness about recent developments in Cost Accounting.
356E	**Cost and Works Accounting. Special Paper III**	1. Understand the basic concepts and principles of cost accounting. 2. Develop budgets for different departments and projects. 3. Estimate the cost reduction and cost control techniques.

Semester VI

Course Code	Course Title	Course Outcome
361	**Business Regulatory Framework-II**	1. Recall the key concepts and principles of business law and regulatory framework. 2. Identify the main laws and regulations that apply to businesses. 3. Explain the purpose and scope of different laws and regulations that impact businesses. 4. Interpret legal texts and case laws to understand the legal implications of business transactions and operations. 5. Apply business laws and regulations to analyze and solve legal issues that arise in business operations. 6. Draft legal documents, such as contracts and agreements, that comply with relevant laws and regulations. 7. Analyze the legal implications of different business models and structures, such as partnerships, corporations, and franchises. 8. Evaluate the impact of regulatory changes on businesses and assess the compliance requirements for these changes. 9. Elaborate the ethical implications of legal and regulatory compliance for businesses, and the impact of non-compliance on stakeholders.
		1. Recall the principles and concepts of advanced accounting and identify the accounting standards and regulatory requirements applicable to advanced accounting topics. 2. Explain the purpose and significance of co-operative society operations their roles , branch accounting, forensic trends.

362	**Advanced Accounting - II**	3. Interpret financial statements that incorporate advanced accounting topics to assess a company's financial position and performance. 4. Apply advanced accounting principles to prepare consolidated financial statements and Analyze the financial impact of transactions and the appropriate accounting treatment. 5. Analyze the financial statements of a multinational company to evaluate performance of the company 6. Identify and analyze the risks associated with complex financial transactions. 7. Evaluate the effectiveness of accounting policies and procedures related to advanced accounting topics in ensuring accurate and reliable financial reporting. 8. Elaborate the ethical implications of financial reporting decisions related to advanced accounting topics, ratio analysis
364	**Auditing & Taxation – II**	1. Demonstrate the Various Concepts Of Income Tax. 2. Understand & Apply to Calculate the Income Of Taxable Salaried Employees 3. Summarize the Total Income and Solved Practical Problems Related to It 4. Adapt the Working of Income Tax Website.
365E	**Cost And Works Accounting. Special Paper IV**	1. Define different methods of costing. 2. Explain and describe the Job Cost Sheet. 3. Solve the problems of contract costing, Process costing and Operating Costing. 4. Elaborate the basic concept of CAS13: service cost centre.

366E	**Cost And Works Accounting. Special Paper V**	1. Define & memorize the meaning and concepts of standard costing and the formulae to solve the practical questions. 2. Discuss the pricing techniques and application of It. 3. Relate the concept of cost accounting standards and how cost treatments can be prepared as per the CAS 4. Estimate the procedure of preparation of cost records and how cost audit can be conducted
363	**International Economics**	1. Contrast the importance of maintaining equilibrium in the balance of payments and suggests suitable measures to correct the disequilibrium 2. Discover and know the causes and effects of deficits in the balance of payments, measures adopted to correct the deficits 3. Analyse the structure of BOP, disequilibrium in BOP, causes of disequilibrium 4. Evaluate the concept of foreign exchange rate and determine its equilibrium exchange rate. 5. Create a comparative analysis table for the working and functions of various International Organisations

CHRIST COLLEGE PUNE

DEPARTMENT OF COMMERCE

Programme Outcome

After the successful completion of two-year MCOM Programme, the graduate will be able to:

Po1: Advanced knowledge of business and commerce: MCom graduates should have a strong understanding of business principles and concepts and be able to apply this knowledge to practical situations

Po2: Specialized skills: Depending on the electives chosen, MCom graduates should have specialized skills in areas like finance, accounting, taxation, or costing

Po3: Critical thinking and problem-solving skills: MCom graduates should be able to analyze complex business situations and develop creative solutions to problems

Po4: Research skills: MCom graduates should have the ability to conduct independent research using appropriate research methodologies

Po5: Ethical and professional behavior: MCom graduates should have a strong understanding of ethical issues in business and commerce and be able to behave professionally in all situations

Po6: Global perspective: MCom graduates should have a broad understanding of global business and commerce, including the challenges and opportunities of operating in an international environment

Master of Commerce

Semester I

Course Code	Course Title	Course Outcome
60111	**Costing Techniques and Responsibility Accounting**	1. Illustrate the Budget Preparation Process. 2. Relate the impact of adverse and favourable variances on cost of a product/service. 3. Explain the industry specific cost ratios. 4. Create the importance of various tools to evaluate the business centre
60109	**Advanced Cost Accounting**	1. Relate the Scope of Cost Accounting in any business Activity. 2. Infer Cost Accounting treatment in relation to Material Cost Accounting, Employee cost and overheads. 3. Develop the learners to establish the interface between Cost Accounting Standards and the various elements of Cost. 4. Build the different methods of costing in Manufacturing and Service industries
60101	**Management Accounting**	1. Apply the concept of management accounting and its significance in the business. 2. Analyze the financial statements. 3. Develop and apply the techniques of management accounting in the financial decision making in the business corporates. 4. Improve the competence with their usage in managerial decision making and control.

60127	**Research Methodology and Analysis**	1. Demonstrate knowledge of research processes 2. Compare the literature reviews using print and online databases (reading, evaluating, and developing 3. Classify American Psychological Association (APA) formats for citations of print and electronic materials 4. Identify, explain, compare, and prepare the key elements of a research proposal/report 5. Elaborate the quantitative and qualitative research.
60102	**Industrial Economics**	1. Develop the factors that influence firm behavior and the strategic decisions firms make, including pricing, output, investment, and entry/exit decisions. 2. Discover market concentration, assess the degree of competition, and evaluate market power to analyze the performance of industries and markets

Semester II

Course Code	Course Title	Course Outcome
60209	**Application of cost accounting**	1. Classify the need to integrate financial and Cost Accounts. 2. Develop understanding about PLC and VCA Concepts. 3. Assess the logic behind ABC technique. 4. Solve cost formats under ABC & to compare such results with the Traditional Overhead Accounting. 5. Develop the importance of Transfer Pricing & Target Costing in the changing scenario.

60201	**Financial Analysis and Control**	1. Recall basic financial analysis concepts, such as financial ratios, valuation techniques, and financial statement analysis tools. 2. Explain the purpose and importance of financial analysis, as well as the role of financial information in decision-making. And comprehend how financial analysis tools can be used to evaluate the financial performance and position of a business. 3. Apply financial analysis techniques to evaluate the financial performance and position of a business entity. They should also be able to use financial analysis software and other tools to process financial data and generate reports. 4. Analyse financial statements and other financial information using various financial analysis techniques to identify strengths, weaknesses, opportunities, and threats of a business entity. 5. Evaluate the financial performance and position of a business entity based on their financial analysis.
60202	**Strategic Management**	1. Understand the ability to think strategically and analyze complex business problems from a strategic perspective. 2. Explore how to conduct thorough internal and external environmental analyses, including SWOT analysis, PESTEL analysis, and industry analysis.

		3. Develop the challenges and techniques involved in implementing strategies throughout an organization, including resource allocation and organizational change.

Semester III

Course Code	Course Title	Course Outcome
321	**Business Finance**	1. Compare with corporate finance required for Indian Industries. 2. Apply the latest developments in the field of corporate finance. 3. Discover the traditional theories of capitalization and dividend distribution practices. 4. Design the working capital management practice of finance to students Skills to be developed.
327	**Cost Audit**	1. Infer the knowledge to the students on Cost Audit Practices. 2. Compare the role and responsibilities of Cost Auditor 3. Develop the students how Cost Audit Report is prepared.
328	**Management Audit**	1. Classify importance of management Audit. 2. Explain the Procedure of Management Audit. 3. Determine the Corporate Image in Management Audit. 4. Interpret the different Areas of Management Audit. 5. Design the Operational Audit

395	**Introduction to Constitution**	1. Explain the philosophy of Constitution of India to students. 2. Discuss them with their freedoms and responsibilities.
322	**Research Methodology**	1. Demonstrate knowledge of research processes 2. Create literature reviews using print and online databases (reading, evaluating, and developing 3. Employ American Psychological Association (APA) formats for citations of print and electronic materials 4. Identify, explain, compare, and prepare the key elements of a research proposal/report 5. Compare and contrast quantitative and qualitative research.
392	**Introduction to Cyber Security**	1. Understand the basics of cybersecurity, including key terminology and principles. 2. Analyse and gain exposure to cybersecurity tools and technologies. 3. Develop hands-on labs and exercises to apply knowledge practically.
394	**Skill Development**	1. Develop proficiency in the targeted skill, whether it's a technical, creative, or soft skill. 2. Enhance critical thinking skills to adapt and innovate while using the skill. 3. Create a portfolio or showcase of work to demonstrate proficiency in the skill.

Semester IV

Course Code	Course Title	Course Outcome
421	**Capital Market and Financial Services**	1. Illustrate the concept of capital markets, types of securities traded, and how the capital market works. 2. Identify and analyse the roles and functions of various financial institutions such as banks, investment banks, brokerage firms, and mutual fund companies 3. Define the regulatory framework governing capital markets and financial services, including the role of regulatory bodies such as securities and exchange commissions and central banks. 4. Develop analytical and critical thinking skills to evaluate financial data, assess risk, and make informed investment decisions.
427	**Recent advances in cost auditing & cost system**	1. Understand the concept of Cost Accounting standard issued by icmai.in 2. Assess & articulate the concept of Excise audit, VAT audit & Productivity audit 3. Analyse how ERP system works in company through field visit. 4. Evaluate the concept of six sigma & its utility
492	**Introduction to Cyber Security IV**	1. Classify advanced cyber threats, such as APTs (Advanced Persistent Threats), zero-day vulnerabilities, and sophisticated attack techniques.

		2. Develop advanced incident response skills, including rapid detection, containment, and recovery from cyber incidents.
422A	**Industrial Economic Environment**	1. Recall and recognize key concepts, theories, and terminology related to industrial economics and its environment. 2. Explain and summarize the principles of industrial economics and its environment, including market structures, pricing strategies, and regulations. 3. Apply economic theories and principles to analyse and solve problems related to industrial economics and its environment, such as market competition and government policies. 4. Analyse and evaluate the impact of various economic factors on industrial environment, such as supply and demand, market structure, and government intervention. 5. Design and evaluate the effectiveness of economic policies and strategies in improving industrial environment and promoting economic growth.
428	**Project Work/Case study**	1. Apply theoretical knowledge gained in previous coursework to practical, real-world situations. 2. Formulate the problem-solving abilities by tackling complex issues or challenges presented by the project or case study.
494	**Skill Development II**	1. Develop a higher level of proficiency in the targeted skill, building upon the basics learned in the prerequisite course. 2. Create critical thinking skills related to the skill, enabling students to analyze complex scenarios or challenges.

		3. Develop building a portfolio or showcase of work that demonstrates advanced proficiency in the skill.

CHRIST COLLEGE PUNE

DEPARTMENT OF MANAGEMENT

Programme Outcome

After the successful completion of three-year BBA Programme, the graduate will be able to:

PO1: Select effective communication skills in a variety of organizational settings

PO2: Develop comprehensive strategic and tactical plans for an organization

PO3: Understand the importance of independent and collaborative inter or multidisciplinary diverse work environment

PO4: Apply problem-solving and decision-making frameworks that propose defensible solutions to organizational opportunities, challenges, changes, and risks

PO5: Evaluate the dynamic of the Global Business Environment from a competitive and economic perspective

PO6: Create a business plan including business feasibility assessment and financial analysis projections for an innovative new business product or services

Bachelor of Business Administration

Semester I

Course Code	Course Title	Course Outcome
101	**Principles of Management**	1. Recall fundamental concepts and principles of management, including the basic roles, skills, and functions of management; 2. Conceptualize how organizations manages its activities and functions.
102	**Business Communication Skills**	1. Explain the implications of effective communication 2. Develop an appropriate understanding role and utility of written communication in life 3. Build proficiency in writing business letters and other communication required in business 4. Analyze proficiency in using various communication media effectively 5. Classify effectively by using different forms of social media
103	**Business Accounting**	1. Develop right understanding regarding role & importance of monetary and financial transactions in business. 2. Cultivate right approach towards classifications and their implications. 3. Evaluate proficiency in preparation of basic financial as to how to write basics accounting treatment- Trading & profit & Loss account statement

104	**Business Economics-Micro**	1. Recall the role of economics as it influences society and business 2. Develop how different decisions are taken in relation in relation to price, demand and supply 3. Decide right understanding regarding, monopoly, perfect completions, revenue etc
105	**Business Mathematics**	1. Develop appropriate understanding as how to use mathematics like computation interest, profit etc 2. Cultivate right understanding regaining numerical aptitude 3. Design logical approach towards analytical approach data
106	**Business Demography**	1. Acquaint the Students with basic concept, Principles and functions of Management. 2. Aware students about the recent trends in Management 3. Examine the students with a comprehensive platform for career development, innovation and further study

Semester II

Course Code	Course Title	Course Outcome
202	**Principles of Marketing**	1. Introduce to the student's how marketing plays in the successful operation of an organization 2. Demonstrate a clear understanding of the marketing concept 3. Evaluate the changes taking place in the macroeconomic environment and its impact on marketing program and accordingly create a marketing plan

		4. Understand key elements of Product Life Cycle along with the various concepts of Marketing Mix
203	**Principles of Finance**	1. Understand the role and importance of Finance in business. 2. Evaluate the sources of finance for raising capital. 3. Analyse the capital structure of the business 4. Compare and evaluate the types of mutual funds 5. Investigate the reasons for under and over-capitalization to take remedial actions
204	**Basics of Cost Accounting**	1. Understand the various concepts of cost accounting 2. Differentiate between types of accounting 3. Analyse the elements of cost and types of costs for cost reduction and cost control 4. Prepare a comprehensive Cost sheet 5. Review the process cost and contract cost of a company
205	**Business Statistics**	1. Introduction to statistics and its objectives 2. Understand the need for classification of raw data 3. Modify raw data into tabulation or organized data 4. Analyze the graphs and diagrams to provide solutions 5. Evaluate the performance of an organization using statistical methods 6. Formulate solutions for the growth and development

206	**Fundamentals of Computers**	1. Describe computer with its characteristics, usage, limitations, and benefits 2. Use computer peripherals and memory efficiently 3. Develop programming concepts 4. Compare between various operating systems and their functions 5. Identify network fundamentals and various communication network
201	**Business Organisations and Systems**	1. Understand the various types of business organizations, including sole proprietorships, partnerships, corporations, and limited liability companies (LLCs). 2. Learn about corporate governance principles, including the roles of boards of directors and executive management in decision-making and oversight. 3. Explore marketing strategies, sales techniques, and customer relationship management (CRM).

Semester III

Course Code	Course Title	Course Outcome
301	**Principles of HRM**	1. Remember the basic concepts of HRM and develop knowledge about its functions. 2. Understand the role of HRP and job analysis in organizational fun toning.

		3. Develop an understanding of Job evaluation, employee morale, job satisfaction and cultivate a right approach towards HRM &their role in business. 4. Demonstrate climb up in career ladder through an understanding of career planning. 5. Create awareness on changing trends in HRM
302	**Supply Chain Management**	1. Enable the students to have a comprehensive understanding of Supply Chain Management. 2. Understand key concepts and issues of Logistics and Inventory Management. 3. Evaluate Warehousing and its role in Space Management
304	**Fundamental of Rural Development**	1. Remember the development issues related to rural society. 2. Analyze the employment opportunities for rural youth. 3. Create interest among the rural youth to participate in rural development programmes and schemes for sustainable development
305	**Management Accounting**	1. Impart basic knowledge of management accounting. 2. Understand the implications of various financial ratios in decision making. 3. Apply and use various tools of management accounting in the business
306	**Banking and Finance**	1. Understand banking function and its operations 2.Evaluate the functioning of Regulatory Authorities in India 3. Create recent technology in banking industry
		1. Describe the major theories, concepts, models and frameworks in the field of Organisational Behaviour.

305 HR	**Organisational Behaviour**	2. Explain determinants of Organisational Behaviour at Individual, Group and Organisational Level. 3. Examine knowledge about approaches to line-up individual, groups & managerial behaviour in order to achieve organisational goals.
306	**Legal aspects in HR**	1. Explain the rights of employees at the workplace. 2. Understand the Applications of differ rent Legal Aspects in HR.
303	**Global Competencies and Personality Development**	1. Develop an understanding of diverse cultures, traditions, and values to foster cultural sensitivity and respect. 2. Enhance communication skills, including verbal, non-verbal, and intercultural communication, to interact effectively in a global context. 3. Learn to work collaboratively in cross-cultural teams, leveraging the strengths of diverse team members.

Semester IV

Course Code	Course Title	Course Outcome
401	**Entrepreneurship and Small Business Management**	1. Understand the role of entrepreneurs in growth and development 2. Predict the business changes and identify the business opportunities the business environment 3. Demonstrate various tools and techniques to solve the problems that occur in the organization

		4. Develop strategies to build the business
402	**Productions And Operations Management**	1. Understand the various manufacturing systems and their applications. 2. Evaluate the different tools to design and develop a product. 3. Apply the various techniques of Quality management and Safety measures. 4. Design a production plan and implement production control
403	**Decision Making and Risk Management**	1. Define different concepts relating to decision making 2. Recommend the best alternative while deciding 3. Analyse an organization's decision-making system
404	**International Business Management**	1. Understand the various concepts of international business and international theories. 2. Differentiate between domestic and international business. 3. Evaluate that how country can gain through international trade practices. 4. Assess what to produce and what to outsource in to the trade. 5. Compare currency exchange rate in the international market. 6. Plan for own international business in future.
405-C-HRM	**HRM Functions & Practices**	1. Integrate the perspective on role of HRM in modern business. 2. Plan the various skills required in human resources and implement them 3. Demonstrate an ability to recruit, train and appraise the employees according to their performance 4. Develop employability skills of the students for the future industry and society requirements

406-C-HRM	**Employee Recruitment & Record Management**	1. Understand the techniques of manpower forecasting, recruitment, and selection. 2. Classify the new methods of re recruitment ie: e-recruitment and talent acquisition. 3. Differentiate between internal and external sources of recruitment. 4. Application of new trends of record management in organisation. 5. Develop Modules for collection and processing of information to generate reports. 6. Designing of computer technologies and its applications in various field with learned concepts
405-B-FM	**Business Taxation**	1. Understand the fundamentals of income tax 2. Apply tax knowledge and calculate and evaluate individual income 3. Develop strategies for text planning to mitigate the tax burden 4. Analyse the regulation relating to advance tax and tax refund 5. File income tax returns 6. Calculate the gross income of an individual
406-B-FM	**Financial Services**	1. Understand the components of the Indian financial system 2. Recognize the role of primary and secondary markets in economic development 3. Compare and evaluate the types of mutual funds 4. Analyse the functioning of the stock markets 5. Examine various financial services provided by financial intermediaries 6. Review the role of information technology in financial services

Semester V

Course Code	Course Title	Course Outcome
502	**Database Administration and Data Mining**	1. Recall the DBMS concepts. 2. Understanding of the working of the transaction management in the DBMS. 3. Develop Data Warehousing and its scope and limitations. 4. Design Data Analytics and Mining and its scope and limitations
501	**Research Methodology**	1. Develop an understanding of the right approach of Research Methodology and its role in Business. 2. Design an understanding of the basic framework of the identification of various sources of information for data collection. 3. Evaluate various Designs, Tools, and Techniques of Research Study. 4. Enable the students in conducting Research work and write Research Paper and Research Project Report
503	**Business Ethics**	1. Understanding of the concepts of Business Ethics 2. Develop theoretical tools to understand current ethical issues and their impacts on business. 3. Analyze the role of ethics in business, Government and Society. 4. Evaluate the Ethical scenario concerning to Environment and consumer protection.
504	**Management of Corporate social Responsibility**	1. Understand the concept and process of CSR 2. Compare the industrial contribution for CSR Policy 3. Examine the context of CSR of present-day Management 4. Discuss the contribution of CSR for the development of Society

505-FM	**Analysis of Finance Statement**	1. Develop the conceptual framework of financial analysis and provide practical exposure to apply various tools of Financial Statement Analysis. 2. Enable to use of various types of ratios for financial and investment decisions. 3. Adapt knowledge about Cash Flow and Fund Flow Statements and their importance in financial analysis
506-FM	**Legal Aspects of Finance and security Law**	1. Identify overview of Indian Financial system 2. Inspect in depth FMC, PFRDA 3. Examine ICDR, listing & delisting of various securities, ESOP, ESOS. 4. Compare companies act 2013 and GST act 2017
505-HRM	**Cross Cultural HR and industrial Relations**	1. Understand the relationship between Cross-Culture Management and Human Resource Management. 2. Develop Cultural Variables in Multinational Enterprises. 3. Construct students with the fundamental knowledge of Industrial Relations.
506-HRM	**Cases in HRM**	1. Understand the application of theory into practice. 2. Design critical thinking by making judgments related to problems in Case Studies of Human Resource. 3. Develop critical thinking for solving Case Studies of Human Resource. 4. Analyze the broad fundamental components of HRM.

Semester VI

Course Code	Course Title	Course Outcome
603	**Business Project Management**	1. Describe the business project management 2. Understanding the difference between projects and programs and categorizing different types of business projects 3. Identifying and managing the risk in business project 4. Analyse the business project management techniques 5. Evaluating a business project and determining the project cost 6. Develop a business project plan using CPM tool
601	**E- Commerce**	1. Describe the concepts of electronic Data Interchange 2. Develop strategies for conducting business through the internet 3. Differentiate between the concepts – Internet, Intranet & Extranet. 4. Explain the role of modern tools used in E-marketing
604	**Management of Innovations and Sustainability**	1. Identify the role of innovation in creating better products, services and processes 2. Describing the relationship of innovation and sustainability 3. Apply different approaches to solve the business problems 4. Explain various challenges that have recently spurred sustainability innovation
602	**Management Information System**	1. Understand the basic concept of MIS and the need for an information system. 2. Relate active participation in an understanding of Information Technology and MIS. 3. Practical Knowledge Acquisition about System Analysis and Design.

		4. Apply a decision-making model on business. 5. Design forms, reports of a system 6. Develop flow charts and data flow diagrams of the system
605-HRM	**Global Human Resource Management**	1. Understand the development and significance of global HRM, recruitment and selection. 2. Differentiate between global HRM and Domestic HRM. 3. Employ the technology in global HRM 4. Evaluate the ethics related challenges for HR functions in multinational enterprises. 5. Arrange training and development sessions for performance management.
606-HRM	**Recent Trends & HR Accounting + Project**	1. Integrate the perspective on role of HRM in modern business. 2. Plan the various skills required in human resources and implement them 3. Demonstrate an ability to recruit, train and appraise the employees according to their performance 4. Develop entrepreneurial skills and employability of the students for the future industry and society requirements
605-FM	**Finance-III Financial Management**	1. Evaluate the sources of finance for raising capital 2.Understand the definition of financial management 3. Analyze the capital structure and cost of capital of a business 4. Investigate the reasons for Under and over-capitalization to take remedial action 5. Assess the different investment proposals to select the best investment alternative 6. Design of financial structure for a business

606	**Finance-IV Case in Finance/Project (Internship)**	1. Evaluate the source of finance for raising capital 2. Analyse the capital structure and cost of capital of a business 3. Assess the different investment proposals to select the best investment alternative 4. Develop the skills to manage working capital

CHRIST COLLEGE PUNE

Department of Management

Programme Outcome

After the successful completion of three-year BBA (CA) Programme, the graduate will be able to:

PO1: Understand the fundamental concepts of computers, software, hardware, and the evolution of computer technologies.

PO2: Emerge with competency in the subject of Business Administration and apply knowledge to cater to the needs of Society / Employer / Institution / Own Business Enterprise

PO3: Imbibe analytical/critical/logical/innovative thinking skills in the field of Business, Management, and Computer Applications

PO4: Analyze, design, implement, and evaluate computerized solutions to real-life problems, using appropriate computing methods including web applications.

PO5: Apply techniques of software validation and reliability analysis to the development of computer programs

PO6: Acquire technical skills to produce industry-ready resources and bring out the true spirit of entrepreneurship.

Bachelor of Business Administration (Computer Applications)

Semester I

Course Code	Course Title	Course Outcome
CA-102	**Principles of Management**	1. Understand fundamental concepts and principles of management, including the basic roles, skills, and functions of management; 2. Conceptualize how organizations manages its activities and functions.
CA-104	**Database Management Systems**	1. Define the Database concepts 2. Analyse the databases and various queries execution 3. Assess and transform information into relational database and produce entity- relationship model for particular system. 4. Design various operations on database
CA-101	**Reasonable knowledge of the Business communication Skills**	1. Understand what is the role of communication in personal and business world 2. Classify system and communication and their utility 3. Develop proficiency in how to write business letters and other communications in required form
CA-103	**C Language**	1. Demonstrate and trace the execution of programs written in C language. 2. Discover the C code for a given algorithm. 3. Evaluate and to implement Programs with pointers and arrays, perform pointer arithmetic, and use the pre-processor. 4. Develop and to perform operations using derived data types.

CA-105	**Statistics**	1. Understand role and importance of statistics in various business situations 2. Assess the skills related with basic statistical technique 3. Develop right understanding regarding regression, correlation and data interpretation

Semester II

Course Code	Course Title	Course Outcome
CA-201	**Organization Behavior & Human Resource Management**	1. Understand the concept of organizational behavior and Human Resource Management 2. Examine the scope of Human Resource Management 3. Identify the role of human resources in helping an organization gain a competitive advantage 4. Analyse the recent trends in training and development 5. Evaluate the factors affecting the recruitment and selection procedure 6. Develop Strategies for overcoming the challenge of cultural diversity in an organization
CA-202	**Financial Accounting**	1. Understand the role of accounting in business 2. Record the financial transaction in books of accounts 3. Prepare final accounts to summarise the financial position of a business 4. Reconcile the cash book and passbook with the help of the bank reconciliation statement

		5. Adapt the accounting software to record and prepare financial statements
CA-203	**Business Mathematics**	1. Demonstrate the appropriate understanding as how to use mathematics like computation interest, profit etc 2. Construct right understanding regaining numerical aptitude 3. Develop logical approach towards analytical approach data
CA-204	**Relational Database**	1. Understand and explain the database concepts like data types, operators and control statements. 2. Apply the database concepts in writing PL/SQL programs. 3. Distinguish between DBMS and RDBMS 4. Select from various programming constructs like cursors, triggers, functions, and procedures and use them while creating programs. 5. Create and organise data in tables and make modifications through PL/SQL programs
CA-205	**Web Technology HTML-JS-CSS**	1. Understand the Structure and implement HTML/CSS. (L2) 2. Apply intermediate and advanced web development practices. (L3) 3. Implement basic JavaScript. (L3) 4. Create visualizations in accordance with UI/UX theories and create webpages that function using external data. (L6) 5. Develop a fully functioning website and deploy on a web server. (L6)

Semester III

Course Code	Course Title	Course Outcome
CA-301	**Digital Marketing**	1. Determine the knowledge about using digital marketing in and as business. 2. Formulate the SWOT analysis, SEO optimization and use of various digital marketing tools.
CA-302	**Data Structure**	1. Understand the concepts of ADTs 2. Illustrate the need for data structures while building applications. 3. Analyse the efficiency and optimize the algorithms 4. Apply advanced linear and non-linear data structures for real world problems 5. Design the Tree and Graph structures
CA-303	**Software Engineering**	1. Demonstrate the System concepts. 2. Assess the Software Engineering concepts. 3.Formulate the software engineering concepts and Design in Software
CA-304	**Angular JS**	1. Understand Client Side MVC and SPA 2. Explore AngularJS Component 3. Develop an AngularJS Single Page Application 4. Create and bind controllers with JavaScript 5. Elaborate the filter in AngularJS application
CA-305	**Big data / Block chain**	1. Infer the expert knowledge and analytical skills in current and developing areas of analysis statistics, and machine learning 2. Identify, develop and apply detailed analytical, creative, problem-solving skills.

		3. Determine the learner with a comprehensive platform for career development, innovation, and further study. 4. Estimate the architecture of R and WEKA with practical's.

Semester IV

Course Code	Course Title	Course Outcome
CA401	**Networking**	1. Discuss the basic of networking concepts. 2. Analyse in detail and understood the basic idea of different protocol. 3. Compare the routing, packet switching and routing algorithms concepts. 4. Interpret the services of connectionless and connection-oriented protocols. 5. Formulate the internet domains and its services
CA-402	**Object Oriented Concept ThroughCPP**	1. Understand various object-oriented concepts. 2. Analyse the different OOPs concepts and apply them to write programs. 3. Compare various dynamic memory management techniques like pointers, constructors, destructors. 4. Interpret the appropriate concepts and use them in writing CPP programs. 5. Create classes and objects which implements all the OOPs features.
		1. Understand the services provided by operating systems. 2. Examine to some details of major OS concepts.

CA-403	**Operating System**	3. Evaluate the issues related to memory management and various related algorithms. 4. Design issues related to file management and various related algorithms 5. Discuss with an operating system via system calls 6. Combine the different types of memory used in OS
CA-404	**Node JS**	1. Understand Node JS and REPL terminal. 2. Experiment with Node JS Modules and Node Package Manager. 3. Find and use code packages based on their documentation to produce working results in a project. 4. Develop applications to handle events in Node JS

Semester V

Course Code	Course Title	Course Outcome
CA-501	**Cyber Security**	1. Understand the Cyber Security and the Tools. 2. Identify the different types of Cyber Crimes. 3. Develop Cyber forensics awareness. 4. Estimate the attacks, security policies and credit card frauds in mobile and Wireless Computing Era
		1. Understand the fundamentals of object modelling 2. Classify and differentiate Unified Process from other approaches. 3. Design with static UML diagrams. 4. Improve with the UML dynamic and implementation diagrams.

CA-502	**Object Oriented Software Engineering (OOSE)**	5. Improve the software design with design patterns. 6. Develop the software against its requirements specification
CA-503	**Core Java**	1. Relate the real-world problems using OOP techniques. 2. Solve problems using java collection framework and I/o classes. 3. Develop multithreaded applications with synchronization. 4. Design GUI based applications
CA-504	**Python**	1. Understand the need and importance of Python language. 2. Examine how to design and implement Python applications. 3. Design and implement a program to solve a real-world problem 4. Formulate the knowledge of handling the concepts of exceptions and files
CA-507	**IOT**	1.Illustrate the role of IoT protocols for efficient network communication. 3. Identify IoT platform such as Arduino Uno. 4. Relate about real time IOT Devices 5. Design the key technologies, smart objects, IoT Architecture and security in Internet of Things.

Semester VI

Course Code	Course Title	Course Outcome
CA-601	**Recent Trends in Information Technology**	1. Understand the basic concepts AI. 2. Apply basic, intermediate and advanced techniques to mine the data. 3. Compare the concept of Spark programming and to Analysis of emergent research data 4. Determine and familiarity with emergent technologies/resources 5. Formulate and integrate principles.
CA-602	**Software Testing**	1. Define the approaches to software testing and identify different testing tools for appropriate environments. 2. Apply the various testing strategies on specialized environments. 3. Evaluate different software metrics using the various measures of complexity 4. Design Test Cases and Create Test Plans based on various requirements under various test scenarios.
CA-603	**Advanced Java**	1. Understand concepts of JDBC Programming, Multithreading and Socket Programming, Spring and Hibernate. 2. Differentiate between servlet-JSP and Spring-Hibernate 3. Develop application by using JDBC,Servlet- JSP and Spring-Hibernate framework
CA-604	**Android Programming**	1. Define app requirements for digital devices. 2. Classify the constraints in developing real time app. 3. Apply skill on real time applications. 4. Evaluate the logic and challenges of Android programming. 5. Develop and Integrate Java and android to develop game and applications for different Industries.

CHRIST COLLEGE PUNE

DEPARTMENT OF MANAGEMENT

Programme Outcome

After the successful completion of three-year BBA (IB) Programme, the graduate will be able to:

PO1: Understand Management discipline with a global emphasis.

PO2: Evaluate and present market research to support an organization's business decision-making.

PO3: Describe and list the documents and the application required in foreign trade.

PO4: Evaluate the impact of the legal environment on an organization's business initiatives.

PO5: Identify and interpret relevant international financial documents, and evaluate the financial strategies of the corporation.

PO6: Develop and present an international business plan

Bachelor of Business Administration (International Business)

Semester I

Course Code	Course Title	Course Outcome
101 GC	**Principles of Management**	1. Understand fundamental concepts and principles of management, including the basic roles, skills, and functions of management; 2. Conceptualize how organizations manages its activities and functions.
102 SC	**Business Communication Skills**	1. Understand the implications of effective communication 2. Develop an appropriate understanding role and utility of written communication in life 3. Develop business letters and other communication required in business
103 GC	**Business Accounting**	1. To develop right understanding regarding role & importance of monetary and financial transactions in business. 2. Cultivate right approach towards classifications and their implications. 3. Evaluate proficiency in preparation of basic financial as to how to how to write basics accounting treatment- Trading & profit & Loss account statement

104 GC	**Business Economics-Micro**	1. Understand role of economics as it influences society and business 2. Study how different decisions are taken in relation in relation to price, demand and supply 3. Demonstrate right understanding regarding, monopoly, perfect completions, revenue etc
105 GC	**Business Mathematics**	1. Develop appropriate understanding as how to use mathematics like computation interest, profit etc 2. Cultivate right understanding regaining numerical aptitude 3. Evaluate logical approach towards analytical approach data
106 SC	**Business Demography**	1. Recall the students with basic concept, Principles and functions of Management. 2. Aware about the recent trends in Management 3. Assess students with a comprehensive platform for career development, innovation and further study

Semester II

Course Code	Course Title	Course Outcome
		1. Introduce to the student's how marketing plays in the successful operation of an organization 2. Demonstrate a clear understanding of the marketing concept

204	**Principles of Marketing**	3. Evaluate the changes taking place in the macroeconomic environment and its impact on marketing program and accordingly create a marketing plan 4. Understand key elements of Product Life Cycle along with the various concepts of Marketing Mix
203	**Commercial Geography**	1. Recognize the difference between GI and trade mark and its applicability in commercial Activity 2. Understand the commercial Activities in different Geographical features and environment referring to domestic as well as world resources. 3. Analyzing on already believed patterns with facts of present times. 4. Application of their knowledge about commercial Geography to the home trade and international business. 5. Create the plan/ presentation of their own product or business with the given resources and Geographical features.
201	**Basics of Cost Accounting**	1. Understand the various concepts of cost accounting 2. Differentiate between types of accounting 3. Analyze the elements of cost and types of costs for cost reduction and cost control 4. Prepare a comprehensive Cost sheet 5. Review the process cost and contract cost of a company

205	**Business Statistics**	1. Understand the need for classification of raw data 2. Modify raw data into tabulation or organized data 3. Analyze the graphs and diagrams to provide solutions 4. Evaluate the performance of an organization using statistical methods 5. Formulate solutions for the growth and development
206	**Fundamentals of Computers**	1. Describe computer with its characteristics, usage, limitations, and benefits 2. Examine computer peripherals and memory efficiently 3. Develop programming concepts 4. Compare between various operating systems and their functions 5. Identify network fundamentals and various communication network
202	**Origin and Development of Global Business**	1. Understand the historical evolution of global business, including early trade routes, colonialism, and the development of multinational corporations. 2. Comprehend key concepts and theories related to globalization, such as the stages of globalization and the impact on economies and societies. 3. Examine cultural diversity and ethical issues that arise in international business interactions, including cultural sensitivity and corporate social responsibility.

Semester III

Course Code	Course Title	Course Outcome
301	**Elements of HRM**	1. Understand the basic concepts of HRM and develop knowledge about its functions. 2. Evaluate the role of HRP and job analysis in organizational fun toning. 3. Develop an understanding of Job evaluation, employee morale, job satisfaction and cultivate a right approach towards HRM &their role in business. 4. Motivate to climb up in career ladder thru n understanding of career planning. 5. Create awareness on changing trends in HRM
A 306	**Supply Chain Management**	1. Enable the students to have a comprehensive understanding of Supply Chain Management. 2. Understand key concepts and issues of Logistics and Inventory Management. 3. Develop Warehousing and its role in Space Management
304	**Production and Operation Management**	1. Understand key concepts and issues of Production and Operations Management 2. Appreciate the concepts of Plant layouts, material handlings & production planning 3. Comprehend key decision areas of operations and analyse data for effective decision making in operations management
302	**Global Competencies and Personality Development**	1. Develop an understanding of diverse cultures, traditions, and values to foster cultural sensitivity and respect.

		2. Enhance communication skills, including verbal, non-verbal, and intercultural communication, to interact effectively in a global context. 3. Develop leadership qualities, including the ability to inspire and lead diverse teams, promote inclusivity, and make ethical decisions.

Semester IV

Course Code	Course Title	Course Outcome
401	**Import Export Procedure**	1. Describe the basics of Import and Export 2. Explain the details concepts of Import and Export 3.: List the procedure to be followed for import and export of goods 4. Introduce various schemes related to foreign trade
402	**Research Methodology**	1. Remember the nature and classification of research design and the types of research 2. Understand history of research, types of research and research design 3. Apply research methods in social and business sciences 4. Identify and discuss the complex issues inherent in selecting a research problem, selecting an appropriate research design, and implementing a research project. 5. Develop an appropriate framework for research studies

403	**Business Ethics**	1. Define business ethics and its types 2. Understanding the role of stakeholders 3. Differentiate between ethics and laws 4. Analyze the impact of Business ethics in its stakeholders 5. Evaluate the unethical practices in business 6. Formulate the solutions to ethical problems 7. Compare the case study of two organizations
404	**Management Information System**	1. Understand the basic concept of MIS and the need for an information system. 2. Relate active participation in an understanding of Information Technology and MIS. 3. Practical Knowledge Acquisition about System Analysis and Design. 4. Apply a decision-making model on business. 5. Design forms, reports of a system 6. Develop flow charts and data flow diagrams of the system
406	**International Warehouse and Supply Chain Management**	1. Define the role of warehouses in modern supply chains 2. Outline ways to improve warehouse layout 3. Compare the types of technological equipment and robotics used in warehouses

Semester V

Course Code	Course Title	Course Outcome
		1. Understand foreign affairs global issues with international business system.

GC 501	**International Relations**	2. Develop various regional groupings their contributions to strengthening of international relations. 3. Compare the various dispute settlement machineries involved in international relations. 4. Evaluate the socio-cultural and trade relations
GC 504	**International Business Law**	1. Understand international law & define evolution of International economic & trade law 2. Develop Transnational corporation (TNC) 3. Compare UNCITRAL & Business Conventions. 4. Explain significance of international dispute settlement machinery in era of globalization 5. Examine international trade & Indian laws related to it.
GC 503	**Business Reporting and Analysis**	1. Develop student's abilities to analyse & interpret various Economic 2. Examine Business decision making. 3. Design reporting pattern followed in corporate sector as a part of MIS.
504	**Foreign Exchange Management**	1. Understand various financial terminologies and transactions involved in the FOREX market. 2. Decide the operations of foreign markets. 3. Demonstrate the principles of currency values
505	**International Marketing Management**	1. Develop Disciplinary Knowledge in International Marketing 2. Encourage entrepreneurial spirit in students; to make them effective in International Business 3. Acquire Analytical reasoning necessary to understand International Marketing.
		1. Examine marketing concepts and phenomenon to current business events In the Industry

506	**Legal Aspects in Marketing Management**	2. Understand ethical challenges related to responsible management of advertising

Semester VI

Course Code	Course Title	Course Outcome
GC 601	**New Venture Creation and StartUps**	1. Understand the fundamentals of how to manage a new venture. 2. Discuss the process of creation of start-ups. 3. Compare various types of feasibility analysis required in new venture creation 4. Design a business plan
602	**International Project Management**	1. Remember the basic concepts of project management 2. Understand organization structure by applying flow of authority and responsibility. 3. Analyse earned value by evaluating a project financially to revise estimates of cost and time 4. Evaluate risks in project management by making resource chart by finding probability of completion of project. 5. Create new project proposals with help of network diagram, critical path and concepts of crashing network
604		1. Understand the structure and strategies of Agribusiness and Agriculture marketing. 2. Recognise the concept and types of contract farming. 3. Assess the role of banks and cooperative institutions in agribusiness.

	Management of agribusiness and agri export	4. Analyze the job of marketing cooperatives. 5. Create the marketing plan and required documents for Agri export
605	**International Service Management**	1. Define the concept related to service 2. Illustrate the concept of Servicer Marketing Triangle 3. Analyze different service sectors & identify the phase of life cycle. 4. Compare quality of services and analyse the gap
606	**Brand Management**	1. Understanding the comprehensive knowledge of brand, theories and concepts, such as brand equity, identity, personality 2. Analysing the Brand Positioning Strategy and how to tackle the competitors. 3. Measuring the impact of Brand Dimensions and Brand Audit 4 Appraise and summarise the various brand strategies used Globally. 5. Collect and design the skills to build a winning brand strategy and activate the brand in a competitive marketplace
603	**Decision Making and Risk Management**	1. Understand various decision-making models, such as rational decision-making, behavioral decision theory, and bounded rationality. 2. Learn how to identify and categorize different types of risks, including financial, operational, strategic, and reputational risks. 3. Develop strategies for making decisions when faced with uncertain or incomplete information

CHRIST COLLEGE PUNE

DEPARTMENT OF SCIENCE

Programme Outcome

After the successful completion of three-year BSC (CS) Programme, the graduate will be able to:

PO1: To develop problem solving abilities using computer

PO2: To build the necessary skill set and analytical abilities for developing computer-based solutions

PO3: To create awareness about process and product standards

PO4: To train students in professional skills related to software industry

Bachelor of Science (Computer Science)

Semester I

Course Code	Course Title	Course Outcome
CS-111	**Problem Solving using Computer and 'C' Programming**	1. Explore algorithmic approaches to problem solving. 2. Develop modular programs using control structures and arrays in 'C'.
CS-112	**Database Management Systems**	1.Solve real world problems using appropriate set, function, and relational models. 2. Design E-R Model for given requirements and convert the same into database tables. 3. Adapt SQL
MTC-111	**Matrix Algebra**	1. Organize large amount of data in the form of matrices 2. Perform basic matrix operations 3. Solve simple linear equations using matrices
MTC-112	**Discrete Mathematics**	1. Identify certain parameters and properties of the given graphs. 2. Design certain algorithms based on Boolean Algebra
ELC-111	**Semiconductor Devices and Basic Electronic Systems**	1. Define and explain electronics components 2. Relate working of semiconductor devices 3. Develop working of Power supply

		4. Examine working of OSCILLATORS 5. Choose appropriate and advanced techniques for data conversion
ELC-112	**Principles of Digital Electronics**	1. Assess interconversion of various number systems. 2. Design and simplify K-maps 3. Evaluate and design encoders and decoders.
CSST-112	**Mathematical Statistics**	1. Solve counting problems related to permutations and combinations using counting principles 2. Assess probability theory in modelling and understanding modern definition of probability. 3. Identify various probability models. 4. Discover numerical problems related to real life situation
CSST-111	**Descriptive Statistics**	1. Learn to describe and summarize data using appropriate measures of central tendency, such as mean, median, and mode. 2. Understand measures of data variability, including range, variance, and standard deviation, to assess the spread of data points. 3. Calculate and interpret percentiles and quartiles to understand data distribution and identify outliers.

Semester II

Course Code	Course Title	Course Outcome
CS-121	**Advanced 'C' Programming**	1. Recall the basic concepts of 'C' Programming 2. Understand code organization with complex data types and structures.

		3. Develop modular programs using control structures, pointers, arrays, strings and structures. 4. Differentiate various types of outputs.
CS-122	**Relational Database Management Systems**	1. Classify and explain the database concepts like data types, operators and control statements. 2. Apply the database concepts in writing PL/SQL programs. 3. Distinguish between DBMS and RDBMS 4. Select from various programming constructs like cursors, triggers, functions and procedures and use them while creating programs. 5. Create and organise data in tables and make modifications through PL/SQL programs
MTC-121	**Linear Algebra**	1. Recall the concept of Vector Spaces . 2. Compute basis of given vector spaces. 3. Solve the system of linear equations using the gaussian elimination method. 4. Represent different isomorphisms of vector spaces using linear transformation. 5. Apply Rank-Nullity Theorem for computations of dimension of vector spaces
		1. Understand the concept of Vector Spaces. 2. Compute basis of given vector spaces.

MTC-122	**Graph Theory**	3. Solve the system of linear equations using the gaussian elimination method. 4. Represent different isomorphisms of vector spaces using linear transformation. 5. Apply Rank-Nullity Theorem for computations of dimension of vector spaces.
ELC-122	**Basics of Computer Organisation**	1. Expalin the working of sequential circuits. 2. Compare and identify the basics of computers. 3. Develop the understanding and identifying the memory architecture. 4. Design their own model using basic knowledge of digital electronic
CSST-121	**Methods Of Applied Statistics**	1. Explain the basic concepts of applied statistics. 2. Analyse the data using correlation and regression 3. Demonstrate the method for solving real life problems using multiple regression and multiple, partial correlation. 4. Formulate real life problems as relevant statistical problems and find the solution using methods of time series

CSST-122	**Continuous Probability Distributions and Testing of Hypothesis**	1. Understand standard continuous probability distributions. 2. Compute probabilities, expected values and variance of standard continuous probability distributions. 3. Apply standard continuous probability distributions to solve real life problems based on given data. 4. Implement testing of hypothesis in decision making situation. 5. Create optimum model against which given data needs to be tested.
ELC-121	**Instrumentation System**	1. Infer the fundamental principles of measurement, including accuracy, precision, sensitivity, and linearity. 2. Familiarize yourself with common measurement instruments, such as voltmeters, ammeters, and oscilloscopes 3. Explore digital communication protocols used in instrumentation systems, such as Modbus and HART.

Semester III

Course Code	Course Title	Course Outcome
CS-231	**Data Structures and Algorithms – I**	1. Interpret organized data structures in PS. 2. Differentiate the usage of various structure in PS. 3. Implementing algorithms to solve problems using DS.
CS-232	**Software Engineering**	1. Compare and chose a process model for a software project development. 2. Identify requirements analyze and prepare models. 3. Prepare the SRS, Design document, Project plan of a given software system

MTC-231	**Group Theory**	1. Differentiate between equivalence relation and congruence relation on Z 2. Identify groups and its elementary properties 3. Explore the properties of finite groups and subgroups
MTC-232	**Numerical Analysis**	1. Develop methods like Newton-Raphson, false position while solving algebraic and transcendental equations. 2. Apply methods like Newton's interpolation, LaGrange's interpolation while solving interpolation problem. 3 Implement methods like Trapezoidal Rule, Simpson's One-Third Rule, Simpson's Three-Eight Rule while finding numerical values of integration 4. Examine methods like Euler's Method, Modified Euler's Method, Runge-Kutta Method while solving ordinary differential equations
ELC-231	**Microcontroller Architecture and programming**	1. Classify programs for 8051 microcontroller 2. Interface I/O peripherals to 8051 microcontroller 3. Design small microcontroller based projects
ELC-232	**Digital Communication and Networking**	1. Define and explain terminologies of data communication 2. Understand the impact and limitations of various digital modulation techniques 3. Acknowledge the need of spread spectrum schemes 4. Identify functions of data link layer and network layer while accessing communication link 5. Choose appropriate and advanced techniques to build the computer network
23921	**Environment Science –I**	1. Expalin the importance of preserving and protecting our Mother Earth. 2. Guide the next generation.

23922	**Language Communication – I**	1. Develop Reading, Writing, Speaking and Listening Skills. 2. Compare certain rules of grammar 3. Develop personality by instilling core values; life skills. 4. Identify soft skills.

Semester IV

Course Code	Course Title	Course Outcome
CS-242	**Computer Networks - I**	1. Recall the basic concept of Networking(LAN,MAN,WAN, Protocols) 2. Develop the OSI and TCP/IP Reference Models and in particular have a good knowledge of lower Layers vs Upper layers and also working of various protocols. 3. Analyze the requirements for a given organizational structure and select the most appropriate networking architecture and technologies.
		1. Understand 2 and 3 dimensional transformations. 2. Compute matrices of two and three dimensional transformations. 3. Apply transformations and projections in computer science.

MTC-241	**Computational Geometry**	4. Represent different equations of planes and curves in three dimensions. 5. Create animation videos using transformations and curves in three dimensions.
MTC-242	**Operations Research**	1. Explain the basic concepts of operations research. 2. Solve LPP using various methods. 3. Demonstrate the method for solving assignment problems and transportation problems. 4. Formulate real life problems as relevant LPP and find the solution using appropriate method.
ELC-241	**Embedded System Design**	1. Differentiate between general computing and embedded systems. 2. Relate the fundamentals of embedded systems. 3. Develop familiarity with tools used to develop in an embedded environment 4. Design their own model using a Single board Computer (Such as Raspberry Pi) for an embedded system application. 5. Design embedded systems and their interfaces with peripheral devices.

ELC-242	**Wireless Communication and Internet of Things**	1. Understand working of wireless technologies such as Mobile communication, GSM, GPRS ,3G and 4G Cellular Network Technologies 2. Apply short-range communication in various applications 3. Experiment above knowledge for data communication 4. Evaluate the concept of Internet of Things 5. Design new IoT based application
24921AECC1	**Environment Science – II**	1. Make use of awareness in the society about the protection of the environment. 2. Create awareness about natural resources, ecosystem, biodiversity and its conservation in urban, rural, industrial, agricultural areas. 3. Discuss issues related to society, human population, environmental pollution etc. 4. Investigate scope of renewable energy
CS-241	**Data Structures & Algorithms - II**	1. Learn about advanced data structures such as heaps, hash tables, balanced trees (e.g., AVL trees, Red-Black trees), and graphs. 2. Understand the principles of dynamic programming and apply it to solve complex problems, including optimization problems and sequence alignment.
24922	**Language Communication II**	1. Develop a higher level of proficiency in the target language, including improved speaking, listening, reading, and writing skills. 2. Achieve greater fluency in spoken language through practice in real-life situations and discussions.

		3. Enhance reading comprehension skills to understand more complex texts and literature.

Semester V

Course Code	Course Title	Course Outcome
CS-351	**Operating Systems - I**	1. Assess concept of operating system and its principle 2. Analyze processes and Thread Scheduling by operating system 3. Create synchronization in process and threads by operating system 4. Design memory management by operating system using with the help of various schemes
CS-352	**Computer Networks – II**	1. Understand the different protocols of Application layer. 2. Develop understanding of technical aspect of Multimedia Systems. 3. Relate various Multimedia Systems applicable in real time. 4. Identify information security goals. 5. Compare and apply cryptographic techniques for data security
CS-353	**Web Technologies – I**	1. How to develop dynamic and interactive Web Page 2. Create well-structured web pages using HTML. 3. Implement user authentication and authorization mechanisms
		1. Perform Exploratory Data Analysis

CS-354	**Foundations of Data Science**	2. Obtain, clean/process, and transform data. 3. Detect and diagnose common data issues, such as missing values, special values, outliers, inconsistencies, and localization. 4. Demonstrate proficiency with statistical analysis of data. 5. Present results using data visualization techniques.
CS-355	**Object Oriented Programming - I (Core Java)**	1. Understand the concept of Object-Oriented Programming such as classes, objects, Packages, and Collections. 2. Develop Web-based applications and GUI-based Applications. 3. Design event-driven Applications.
CS-356	**Theoretical Computer Science and Compiler Construction - I**	1. Recall the Finite Automata, Pushdown Automata and Turing Machine. 2. Evaluate the Regular Language, Context Free Language, Context Sensitive Language and Unrestricted Language. 3. Explain the relation between Automaton and Language
CS-3510	**Python Programming**	1. Develop logic of problem-solving using python 2. Evaluate various Data Structures like List 3. Familiarize with Basic control structures, loops & tune? 4. Develop small Application Project.
CS-3511	**Blockchain Technology**	1. Learn Fundamentals of Blockchain Technology. 2. Learn Blockchain Programming. 3. Demonstrate basic knowledge of smart contracts & how they function

Semester VI

Course Code	Course Title	Course Outcome
CS-361	**Operating Systems – II**	1. Compare deadlock concepts and file handling techniques. 2. Apply disk scheduling algorithms in process management 3. Analyse the working of distributed and mobile O.S.
CS-362	**Software Testing**	1. Define various software testing methods and strategies. 2. Compare a variety of software metrics and latest testing methods used in the software industries. 3. Identify defects and manage those defects for improvement in quality for given software. 4. Design test cases and test plans, review reports of testing for qualitative software.
CS-363	**Web Technologies – II**	1. Recall the basic concepts of HTML, CSS, PHP 2. Remember basic concepts of JavaScript, XML, jQuery, cookies, session, Ajax, CodeIgniter 3. Analyse the difference between server-side Application and client-side Application

		4. Create dynamic website. Using MVC based framework, students will make design and handle the errors in dynamic Website
CS-364	**Data Analytics**	1. Utilize knowledge and skills to continue learning and adapting to new data science technologies. 2. Acquire the fundamental understanding of the analytical techniques and software tools necessary to effectively generate useful information from structured and unstructured datasets of any size. 3. Demonstrates the ability to technical skills in predictive and prescriptive modelling to support decision making based on data analytics. 4. Evaluate to the basics of programming and analytics software used to the solution of real-world business problems.
CS-365	**Object Oriented Programming using Java – II**	1. Recall basic concepts of Database programming, Collection Framework, Multithreading, and dynamic web pages using Servlets and JSP and also basics Spring Framework. 2. Analyze the difference between Java Application and Web Application 3. Create application using Multithreading concepts, database application using JDBC and web application using Servlets and Spring Framework.

CS-3610	**Software Testing Tools**	1. Define various software testing methods and strategies. 2. Evaluate a variety of software metrics and latest testing methods used in the software industries. 3. Identify defects and manage those defects for improvement in quality for given software. 4. Design test cases and test plans, review reports of testing for qualitative software
CS-366	**Compiler Construction**	1. Remember of the overall structure and components of a compiler, including lexical analysis, parsing, semantic analysis, optimization, and code generation. 2. Explore semantic analysis to check and enforce the correctness of program semantics, including type checking and error handling. 3. Develop techniques for managing symbol tables, which store information about variables, functions, and other program entities. 4. Design and implement test cases and techniques for thoroughly testing the compiler to ensure its correctness and robustness.

CHRIST COLLEGE PUNE

DEPARTMENT OF SCIENCE

Programme Outcome

After the successful completion of two-year MSC (CS) Programme, the graduate will be able to:

PO1: Recognize the characteristics of patterns that make it useful to solve real-world problems.

PO2: Process available data using python libraries and predict outcomes using Machine Learning algorithms to solve given problem.

PO3: Able to use specific frameworks as per applications need.

PO4: Design java application using design pattern techniques.

Master of Science (Computer Science)

Semester I

Course Code	Course Title	Course Outcome
CSUT111	**Paradigm of Programming Language**	1. Analyze syntax from semantics 2. Compare programming language designs 3. Criticize their strengths and weaknesses 4. Combine new languages more quickly
CSUT113	**Database Technologies**	1. Distinguish types of NoSQL databases (Document oriented, key Value pairs, Column-oriented and Graph) 2. Evaluate detailed architecture, define objects, load data, query data and performance tune NoSQL databases. 3. Create large volumes of structured, semi-structured, and unstructured data using database technologies.
CSDT114B	**Artificial Intelligence**	1. Examine various types of algorithms in AI. 2. Convey the ideas in AI research & program. 3. Modify numerous applications & huge possibilities in the field of AI that goes Beyond normal human imagination.
CSUT112	**Design and Analysis of Algorithms**	1. Develop the ability to solve complex computational problems using efficient algorithms. 2. Measure various algorithm design paradigms, such as divide and conquer, dynamic programming, greedy algorithms, and backtracking.

		3. Predict various data structures and understand how to choose the most appropriate ones for different algorithmic problems.

Semester II

Course Code	Course Title	Course Outcome
CSUT121	**Advanced Operating System**	1. Identify and use UNIX/Linux utilities to create and manage simple file processing operations, organize directory structures with appropriate security, and develop shell scripts to perform more complex tasks. 2. Create processes background and foreground by process fork and signals system calls. 3. Analyze a given problem and apply requisite facets of SHELL programming in order to devise a SHELL script to solve the problem 4. Demonstrates shared memory segments, pipes, message queues and can exercise inter process communication.

CSUT122	**Mobile Technologies**	1. Understand basic concepts of mobile OS. 2. Deduct simple GUI applications, use built-in widgets and components, work with the database to store data locally, and much more. 3. Demonstrate their understanding of the fundamentals of Android operating systems 4. Explain their skills of using Android software development tools 5. Differentiate between Android OS and Ios 6. Create an Ios user interfaces and application.
CSUT123	**Software Project Management**	1. Understand different components of IT projects. 2. Apply project Management practices to develop strategies to calculate risk factors, to control the design implementation and evaluation of IT Projects, to launch new products, services and events relative to the needs of stakeholders. 3. Develop project plan, Estimate and adjust Project Variables.
CSDT124	**Soft Computing**	1. Gain an understanding of the fundamental concepts and principles of soft computing paradigms, including fuzzy logic, neural networks, genetic algorithms, and swarm intelligence.

		2. Explore artificial neural networks, including feedforward and recurrent architectures, and understand their applications in pattern recognition, classification, and regression. 3. Design and analyze fuzzy logic-based control systems for applications in robotics, automation, and process control.

Semester III

Course Code	Course Title	Course Outcome
CSUT231	**Software Architecture and Design Patterns**	1. Recognize major Software Architectural styles, Designs and Frameworks to create reusable and flexible software applications and systems. 2. Comapre the characteristics of patterns that make it useful to solve real-world problems. 3. Apply Software Architectural styles, Design Pattern and Frameworks to create reusable and flexible software applications and systems. 4. Perform software verification and validation using inspections, design and execution of system. 5. Compare case study to take a framework and find patterns in the frames. 6. Design java applications using design pattern techniques.

CSUT232	**Machine Learning**	1. Recognize the characteristics of machine learning that make it useful to real-world problems. 2. Process available data using python libraries and predict outcomes using Machine Learning algorithms to solve given problem. 3. Estimate Machine Learning models efficiency using suitable metrics. 4. Design application using machine learning techniques.
CSUT233	**Web Frameworks**	1. Relate with the technology which is used widely in Industry as a part of full stack developer. 2. Develop the web application in Python. 3. Assess what really the asynchronous programming. 4. Build and deploy robust Django Web App. 5. Formulate with Restful web services

Semester IV

Course Code	Course Title	Course Outcome
CSUIT241	**Industrial Training / Institutional Project**	1. Compare real-world industrial or institutional setting, applying theoretical knowledge to practical situations. 2. Develop and enhance a range of professional skills, including problem-solving, communication, teamwork, time management, and adaptability.

		3. Acquire an in-depth understanding of the industry or sector in which the training or project is conducted, including its challenges, trends, and best practices.